

Project Initialization and Planning Phase

Date	27 June2026
TeamID	
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal(Proposed Solution) report

The proposal report aims to develop a machine learning-based loan approval prediction system that evaluates applicant creditworthiness and assists banks and financial institutions in making faster and data-driven loan approval decisions. The system analyzes applicant financial and personal information, predicts loan eligibility, and provides prediction results through a Flask-based web application.

Project Overview	
Objective	To develop a machine learning-based loan prediction system that accurately predicts loan approval probability using applicant details such as income, education, employment status, loan amount, credit history, and property area. The system helps financial institutions improve credit evaluation efficiency and reduce manual processing time.
Scope	The project focuses on collecting and analyzing historical loan application data, performing data preprocessing, training multiple classification models, selecting the best-performing model, and deploying the model through a Flask web application for real-time loan eligibility prediction.
Problem Statement	
Description	Traditional loan approval processes require manual verification and analysis, which can be time-consuming and may lead to delays in decision-making. Incorrect credit risk assessment can result in loan defaults and financial losses for banking institutions.
Impact	An accurate loan prediction system helps banks and financial institutions evaluate applicants efficiently, reduce processing time, identify potential risks, improve decision-making, and provide faster loan eligibility assessment.
Proposed Solution	
Approach	Use machine learning classification algorithms to analyze historical loan applicant data, preprocess the data, train multiple models, compare their performance, and deploy the best-performing model using Flask for real-time loan approval prediction.

Key Features	• Machine learning-based loan approval prediction • Historical loan dataset analysis and visualization • Data preprocessing including missing value handling, encoding, scaling, and balancing • Classification models: • Decision Tree • Random Forest • K-Nearest Neighbors (KNN) • XGBoost • XGBoost-based final prediction model • Flask web application for user-friendly loan eligibility assessment • Prediction result with confidence score
--------------	--

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications for model development	Intel i3 or above processor (GPU not required)
Memory	RAM specifications	Minimum 4GB RAM (8GB recommended)
Storage	Space required for dataset, models, and application files	Minimum 10GB free storage
Software		
Frameworks	Python frameworks	Flask
Libraries	Machine learning and data preprocessing libraries	Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Imbalanced-learn, Joblib
Development Environment	IDE/code editor	VS Code / PyCharm
Data		

Data	Source, size, format	Loan Prediction Dataset (loan_prediction.csv) containing applicant details and loan status information
------	----------------------	--

